ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

MARKING SCHEME

JUNE 2016

CHEMISTRY 9189/2

1 (a) (i) $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^{10}$ $4p^1$/ [Ar] $4s^2$ $3d^{10}$ $4p^1$; [1]

~~[1]~~

~~(ii) mass numbers / nucleon number;~~ [1]

(iii) isotopes;

(iv) $x + y = 100$ --------- equation 1 [1]

$$\frac{69x}{100} + \frac{71y}{100} = 69.8 \text{ ----- equation 2}$$

Solve equation by substitution $x = 100 - y$

Substitute in equation 2 [1]

$69(100 - y) + 71y = 6\,980$ -------- equation 3

$71y - 69y = 6\,980 - 6\,900$

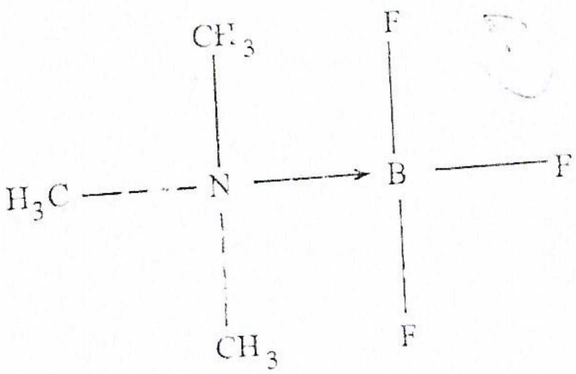
$2y = 80$

$y = 40\%$

$\therefore \quad x = (100 - 40)\%$ [1]

$= \quad 60\%$ [Total: 8]

2 (a) (i)



[1]

[1]

(ii) Dative bond / co-ordinate bond;

(b) (i) $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ [1]

$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$;

(ii) A solution of 0.1 M HC*l* is fully ionised and the number of H_3O^+/H^+ ions is greater; [1]

than in ethanoic acid which is partially ionised; [1]

(c) (i) A buffer solution resits changes in pH when small amount of acid or alkali is added; [1]

(ii) $$\mathrm{pH} = \log_{10} Ka - \log_{10}\frac{[NH_4^+]}{[NH_3]}$$

$$10.0 = -\log_{10} 6\times10^{-10} - \log_{10}\frac{[NH_4^+]}{[NH_3]};$$ [1]

$$Log_{10}\frac{[NH_4^+]}{[NH_3]} = -0.18$$ [1]

$$\frac{[NH_4^+]}{[NH_3]} = 0.16;$$ [1]

[Total: 8]

(a) (i) chlorine is a gas and when it escapes/diffuses, [1]
from solution the equilibrium moves to the left to produce more chlorine; [1]

(ii) alkali neutralises HCl and $HOCl$; [1]
equilibrium moves to the right producing more HCl and $HOCl$ thus reduce chlorine present; [1]

(b) (i) Cd has a higher M.P because of a giant covalent bonded structure; [1]
Pb has a low M.P because of a metallic structure; [1]

(ii) PbO; [1]
CO_2; [1]

(iii) CO_2 – acidic oxide; [1]
PbO – basic oxide; [1]

(iv) stability
+4 oxidation state decreases down the group, [1]
due to inert pair effect. [1]

[Total: 12]

4 (a) (i)

	C	H	O	
% by mass;	66.6	11.2	100-77.8	
Ar	12	1	16	
Ratio	5.55	11.2	1.38	[1]
	4	8	1	

Empirical formula or Q C_4H_8O; [1]

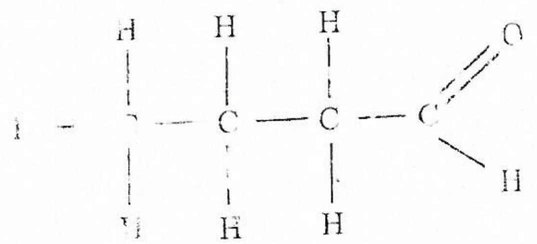
(ii) $(C_4H_8O)n = 72$

$72n = 72$

$n = 1$

$\therefore$ Q is C_4H_8O; [1]

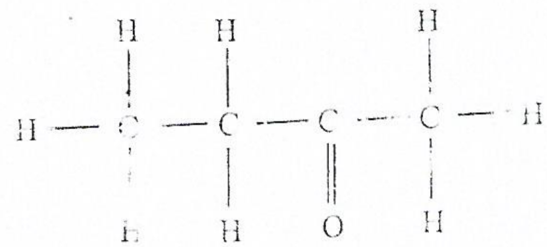
(b) (i) ~~Carbonyl group;~~ [1]



[1]

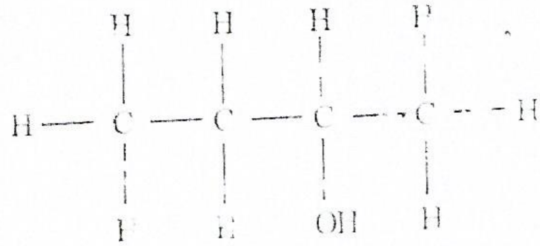
Q is an aldehyde because of the red brown ppt produced with Fehling's solution; [1]

(ii) Structure of R



[1]

R is a ketone because it forms a yellow ppt with triiodomethane test [1]



[1]

butan-2-ol; [1]

	(iv) reduction;			[1]
				[Total: 11]

5 (a)	Reason	Reagent(s)	Condition(s)	
	x	excess concH_2SO_4;	heat at 170°C;	[1]
		[illegible]	[illegible]	[1]
	y	$H+/K_2Cr_2O_7$;	warm;	[1]
		[illegible]		[1]
(b)	A	CH_3COH;		[1]
	B	$CH_3CH_2\cancel{X}$ Cl, Br, I		[1]
	C	$CH_3CH_2NH_2$;		[1]
				[Total: 7]